

MAXIMUM LIKELIHOOD ESTIMATION and MAP

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0.1 Abstract

This report provides a comprehensive examination of two fundamental statistical estimation techniques: Maximum Likelihood Estimation (MLE) and Maximum a Posteriori Estimation (MAP). These methods play a pivotal role in various fields, such as statistics, machine learning, and Bayesian inference, by enabling parameter estimation and decision-making based on observed data.

The report begins by introducing the theoretical foundations of MLE and MAP, elucidating their underlying principles and mathematical formulations. It explores the essential differences between the two methods, with a focus on their respective objectives and approaches to handling uncertainty. Finally, the report concludes with a summary of key takeaways and recommendations for choosing between MLE and MAP in different scenarios. It emphasizes the importance of understanding the underlying assumptions and goals of estimation tasks to make informed decisions.

Overall, this report provides a valuable resource for researchers, practitioners, and students seeking a comprehensive understanding of Maximum Likelihood Estimation and Maximum a Posteriori Estimation, enabling them to make informed choices when faced with statistical estimation challenges in their respective fields.

Contents

0.1	Abstract	1
0.2	Introduction	3
0.3	Maximum Likelihood Estimation(MLE)	4
0.3.1	Definition	4
0.3.2	Overview	4
0.3.3	Key Characteristics	5
0.3.4	Applications	5
0.4	Maximum a Posteriori (MAP)	5
0.4.1	Definition	5
0.4.2	Overview	5
0.4.3	Key Characteristics	6
0.4.4	Applications	6
0.5	Differences between MLE and MAP	7
0.6	Conclusion	7
0.7	References	7

0.2 Introduction

Introduction:

Statistical estimation is a fundamental component of data analysis and inference, serving as the bedrock upon which countless scientific, engineering, and decision-making endeavors are built. Among the multitude of estimation techniques, Maximum Likelihood Estimation (MLE) and Maximum a Posteriori Estimation (MAP) stand out as pivotal methodologies that have shaped the landscape of modern statistics and machine learning. In this report, we embark on a comprehensive exploration of these two estimation techniques, shedding light on their theoretical foundations, practical applications, comparative strengths and weaknesses, and their evolving role in contemporary research.

Maximum Likelihood Estimation, abbreviated as MLE, is a principle deeply rooted in the frequentist paradigm of statistics. MLE seeks to find the values of model parameters that maximize the likelihood function, effectively identifying the parameter values most consistent with the observed data. With its origins dating back to the early 20th century, MLE has established itself as a cornerstone in fields ranging from physics and biology to economics and engineering. It has a reputation for its simplicity and consistency, making it a go-to choice for parameter estimation in many scenarios.

On the other hand, Maximum a Posteriori Estimation, denoted as MAP, is firmly grounded in the Bayesian framework, which embraces uncertainty and incorporates prior knowledge into the estimation process. MAP estimation seeks to find the most probable values of model parameters by considering both the likelihood of the observed data and a prior distribution over the parameters themselves. This Bayesian approach allows for a more flexible and comprehensive treatment of uncertainty, making MAP a potent tool for decision-making, model fitting, and parameter estimation in complex, real-world situations.

0.3 Maximum Likelihood Estimation(MLE)

0.3.1 Definition

Maximum Likelihood Estimation (MLE) is a statistical method used to estimate the parameters of a probability distribution that best describe a given dataset. The fundamental idea behind MLE is to find the values of the parameters that maximize the likelihood of the observed data, assuming that the data are generated by the specified distribution.

0.3.2 Overview

One solution to probability density estimation is referred to as Maximum Likelihood Estimation, or MLE for short.

Maximum Likelihood Estimation involves treating the problem as an optimization or search problem, where we seek a set of parameters that results in the best fit for the joint probability of the data sample (X).

First, it involves defining a parameter called theta that defines both the choice of the probability density function and the parameters of that distribution. It may be a vector of numerical values whose values change smoothly and map to different probability distributions and their parameters.

In Maximum Likelihood Estimation, we wish to maximize the probability of observing the data from the joint probability distribution given a specific probability distribution and its parameters, stated formally as:

$P(X \mid \theta)$ This conditional probability is often stated using the semicolon (;) notation instead of the bar notation ($\mid$) because theta is not a random variable, but instead an unknown parameter. For example:

$P(X ; \theta)$ or

$P(x_1, x_2, x_3, \dots, x_n ; \theta)$ This resulting conditional probability is referred to as the likelihood of observing the data given the model parameters and written using the notation $L()$ to denote the likelihood function. For example:

$L(X ; \theta)$ The objective of Maximum Likelihood Estimation is to find the set of parameters (theta) that maximize the likelihood function, e.g. result in the largest likelihood value.

maximize $L(X ; \theta)$ We can unpack the conditional probability calculated by the likelihood function.

Given that the sample is comprised of n examples, we can frame this as the joint probability of the observed data samples $x_1, x_2, x_3, \dots, x_n$ in X given the probability distribution parameters (theta).

$L(x_1, x_2, x_3, \dots, x_n ; \theta)$ The joint probability distribution can be restated as the multiplication of the conditional probability for observing each example given the distribution parameters.

product i to n $P(x_i ; \theta)$ Multiplying many small probabilities together can be numerically unstable in practice, therefore, it is common to restate this problem as the sum of the log conditional probabilities of observing each example given the model parameters.

$\sum_{i=1}^n \log(P(x_i; \theta))$ Where $\log$ with base-e called the natural logarithm is commonly used.

0.3.3 Key Characteristics

1.Unbiased: MLE estimates are asymptotically unbiased, meaning that as the sample size increases, the estimates tend to converge to the true parameter values.

2.Efficiency: MLE estimators are often efficient, which means they have the smallest possible variance among unbiased estimators.

3.Consistency: MLE estimators are consistent, meaning that as the sample size increases, they converge in probability to the true parameter values.

4.Applicability: MLE can be applied to a wide range of statistical models, including linear regression, logistic regression, and many more.

5.Assumes Uniform Prior: MLE does not incorporate prior information about parameter values.

0.3.4 Applications

1.MLE is commonly used in the estimation of parameters for various probability distributions, including Gaussian, Poisson, and Binomial distributions.

2.In machine learning, MLE is used for parameter estimation in models such as linear regression, logistic regression, and Gaussian mixture models.

0.4 Maximum a Posteriori (MAP)

0.4.1 Definition

In Bayesian statistics, a maximum a posteriori probability (MAP) estimate is an estimate of an unknown quantity, that equals the mode of the posterior distribution. The MAP can be used to obtain a point estimate of an unobserved quantity on the basis of empirical data. It is closely related to the method of maximum likelihood (ML) estimation, but employs an augmented optimization objective which incorporates a prior distribution (that quantifies the additional information available through prior knowledge of a related event) over the quantity one wants to estimate. MAP estimation can therefore be seen as a regularization of maximum likelihood estimation.

0.4.2 Overview

It involves calculating the conditional probability of one outcome given another outcome, using the inverse of this relationship, stated as follows:

$P(A|B) = (P(B|A) * P(A)) / P(B)$ The quantity that we are calculating is typically referred to as the posterior probability of A given B and $P(A)$ is referred to as the prior probability of A.

The normalizing constant of $P(B)$ can be removed, and the posterior can be shown to be proportional to the probability of B given A multiplied by the prior.

$P(A|B)$ is proportional to $P(B|A) * P(A)$ Or, simply:

$P(A \mid B) = P(B \mid A) * P(A)$ This is a helpful simplification as we are not interested in estimating a probability, but instead in optimizing a quantity. A proportional quantity is good enough for this purpose.

We can now relate this calculation to our desire to estimate a distribution and parameters (θ) that best explains our dataset (X), as we described in the previous section. This can be stated as:

$P(\theta \mid X) = P(X \mid \theta) * P(\theta)$ Maximizing this quantity over a range of θ solves an optimization problem for estimating the central tendency of the posterior probability (e.g. the model of the distribution). As such, this technique is referred to as “maximum a posteriori estimation,” or MAP estimation for short, and sometimes simply “maximum posterior estimation.”

maximize $P(X \mid \theta) * P(\theta)$ We are typically not calculating the full posterior probability distribution, and in fact, this may not be tractable for many problems of interest.

Note: this is very similar to Maximum Likelihood Estimation, with the addition of the prior probability over the distribution and parameters.

In fact, if we assume that all values of θ are equally likely because we don’t have any prior information (e.g. a uniform prior), then both calculations are equivalent.

Because of this equivalence, both MLE and MAP often converge to the same optimization problem for many machine learning algorithms. This is not always the case; if the calculation of the MLE and MAP optimization problem differ, the MLE and MAP solution found for an algorithm may also differ.

0.4.3 Key Characteristics

1.Incorporates Prior Information: MAP estimation incorporates prior beliefs or information about the parameters through the prior distribution, which can help stabilize estimates, especially when data is limited.

2.Bayesian Framework: MAP estimation is firmly rooted in the Bayesian framework, making it suitable for cases where prior knowledge is available and important.

3.Regularization: In cases where the prior distribution is informative, MAP can act as a form of regularization, preventing overfitting.

4.Computationally Intensive: Calculating the MAP estimate can be computationally intensive, especially when dealing with complex models and high-dimensional parameter spaces.

5.Subjective Choice of Prior: The choice of the prior distribution can impact the MAP estimate, and different prior choices can lead to different results.

0.4.4 Applications

1.MAP estimation is particularly useful when incorporating prior knowledge or beliefs about the parameters.

2.In Bayesian inference, MAP estimation is essential for various applications, including parameter estimation in hierarchical models and Bayesian neural networks.

0.5 Differences between MLE and MAP

1.Philosophy: MLE focuses solely on finding the parameter values that maximize the likelihood of observed data. MAP combines the likelihood and prior information to find parameter values that are both likely given the data and consistent with prior beliefs.

2.Incorporating Prior Information: MLE does not incorporate prior information about parameter values. MAP explicitly incorporates prior beliefs through the prior distribution.

3.Uncertainty: MLE provides point estimates of parameters, which do not capture uncertainty. MAP estimates can capture uncertainty through the entire posterior distribution

0.6 Conclusion

In conclusion, Maximum Likelihood Estimation (MLE) and Maximum A Posteriori (MAP) estimation are fundamental statistical techniques that play pivotal roles in various fields, particularly in statistical modeling and machine learning. These methods offer powerful tools for estimating parameters in probabilistic models, allowing us to make informed decisions and predictions based on data.

Maximum Likelihood Estimation focuses on finding parameter values that maximize the likelihood of observing the given data. It provides efficient and unbiased parameter estimates and serves as the foundation for many statistical and machine learning algorithms. MLE's simplicity and widespread applicability make it an essential tool in the data scientist's toolkit.

On the other hand, Maximum A Posteriori estimation extends the MLE framework by incorporating prior information about parameter values. This Bayesian approach allows us to make more robust and regularized estimates, particularly when dealing with limited data or when prior knowledge is available. MAP estimation balances the likelihood of the data with our prior beliefs, providing a well-rounded solution for parameter estimation.

0.7 References

1. Casella G., Berger, R. L. (2002). Statistical Inference. Duxbury Press. This book provides a comprehensive introduction to statistical inference, including chapters on MLE and Bayesian methods, which are relevant to understanding MLE and MAP estimation.

2.Bishop, C. M. (2006). "Pattern Recognition and Machine Learning." Springer. This textbook includes a thorough discussion of probabilistic graphical models and Bayesian estimation, which are closely related to MAP estimation.

3.<https://machinelearningmastery.com/maximum-a-posteriori-estimation/>

4.<https://machinelearningmastery.com/what-is-maximum-likelihood-estimation-in-machine-learning/>